SMART STORAGE TANK LEVEL & USAGE PREDICTION SYSTEM

For Petrochemical Plants

Milestone 3 — Live Data Simulation & Model Integration

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1. Milestone Overview

Milestone 3 integrates the machine learning pipeline established in Milestone 2 with live sensor data streamed from the IoT hardware deployed in Milestone 1. The milestone demonstrates a complete end-to-end data flow: from physical ultrasonic sensors mounted on petrochemical storage tanks, through the ESP32-S3 microcontroller and ThingSpeak cloud platform, into a Python-based processing and inference pipeline running in Google Colab.

A key architectural upgrade in this milestone is the transition from height-based to volume-based tank calculations. All sensor distance readings are converted to liquid volume in litres using the cylindrical tank geometry formula ($V = \pi r^2 h$), with tank-specific parameters of radius = 2 m and height = 4 m, yielding a maximum volume of approximately 50,265 litres per tank. A noise filtering and spike rejection pipeline, consistent with the firmware logic running on the ESP32-S3, is applied in Python before any ML inference is performed.

1.1 Milestone Objectives

- Establish cylindrical volume conversion from ultrasonic sensor distance readings.
- Configure and connect to the live ThingSpeak API (Channel 3327870) to retrieve real ESP32 sensor data.
- Implement Python-side signal processing: range validation, spike rejection, and weighted smoothing.
- Process raw ThingSpeak fields into a feature matrix compatible with the Milestone 2 ML pipeline.
- Run the trained Random Forest Regressor and Classifier on live data to predict fill level and risk category.
- Visualise live predictions across six dashboard panels.
- Export a structured prediction CSV for downstream Power BI integration in Milestone 4.

2. Updated System Architecture

Milestone 3 represents the fully integrated state of the system. The data pipeline now operates across all five architectural layers defined in the project specification, with volume-based calculations replacing the earlier height-based approach throughout.

2.1 End-to-End Data Flow

Stage	Component	Description
1	HC-SR04 Sensors	Measure distance from sensor (mounted at tank top) to liquid surface.
2	ESP32-S3	Converts distance to liquid height, then computes volume using $\pi r^2 h$. Uploads to ThingSpeak every 15 seconds.
3	ThingSpeak Cloud	Stores 8 fields per entry: 3 tank levels, 3 water sensor readings, total usage, and time to empty.
4	Python Fetch Layer	Retrieves the latest 200 records via ThingSpeak REST API. Parses and validates JSON response.
5	Signal Processing	Applies range validation, spike rejection, and weighted smoothing to clean sensor readings before volume conversion.
6	Volume Conversion	Converts cleaned distance readings to precise volume in litres using cylindrical geometry.
7	Feature Engineering	Encodes temporal and categorical features using the Milestone 2 label encoders. Scales using the Milestone 2 StandardScaler.

8	ML Inference	Random Forest Regressor predicts fill level in litres. Random Forest Classifier assigns CRITICAL / WARNING / NORMAL risk label.
9	Output & Export	Prediction results visualised in a 6-panel dashboard and exported to milestone3_predictions.csv for Power BI.

3. Cylindrical Tank Geometry & Volume Conversion

A fundamental upgrade in Milestone 3 is the adoption of physics-based volume calculation. Rather than working with sensor-derived height percentages, the system computes the actual liquid volume in litres using the standard cylindrical volume formula. This ensures all predictions and alerts are expressed in operationally meaningful units.

3.1 Tank Physical Parameters

Parameter	Value	Unit
Tank Radius (r)	2.0	metres
Tank Height (h)	4.0	metres (400 cm)
Geometric Full Volume ($\pi r^2 h$)	50,265.5	litres
Operational Capacity Cap	50,265.5	litres (geometric value used)
Litres per cm of height	125.66	L/cm

3.2 Volume Conversion Formula

The core conversion function takes the ultrasonic distance reading (headspace from sensor to liquid surface) and returns the liquid volume in litres:

```
fill_height_cm  = TANK_HEIGHT_CM - distance_cm
fill_height_m   = fill_height_cm / 100
volume_m3       = π × r² × fill_height_m
volume_litres   = volume_m3 × 1000   (clamped to [0, TANK_CAPACITY_L])
```

3.3 Distance-to-Volume Sanity Check

The following table confirms correct conversion across the full sensor range:

Distance (cm)	Liquid Height (cm)	Volume (L)	Fill %
0	400	50,265.5	100.0%
100	300	37,699.1	75.0%
200	200	25,132.7	50.0%
300	100	12,566.4	25.0%
400	0	0.0	0.0%

4. ThingSpeak API Integration

Live sensor data is retrieved directly from the ThingSpeak cloud platform using the REST API. The channel was established in Milestone 1 and continuously receives data from the ESP32-S3 hardware at 15-second intervals.

4.1 Channel Configuration

Parameter	Value
Channel ID	3327870
Base URL	https://api.thingspeak.com
Access	Public channel (no API key required)
Records fetched	200 latest entries per call
Upload interval	15 seconds (from ESP32 firmware)

4.2 ThingSpeak Field Mapping

Field	Variable	Description
field1	tank_1_distance_cm	HC-SR04 distance reading for Tank 1 (cm)
field2	water_sensor_1	Analog ADC value from water level sensor, Tank 1
field3	tank_2_distance_cm	HC-SR04 distance reading for Tank 2 (cm)
field4	water_sensor_2	Analog ADC value from water level sensor, Tank 2
field5	tank_3_distance_cm	HC-SR04 distance reading for Tank 3 (cm)
field6	water_sensor_3	Analog ADC value from water level sensor, Tank 3
field7	total_usage_raw	Total liquid consumed since last reading cycle (litres)
field8	time_to_empty_raw	ESP32-computed predicted time to empty (hours)

5. Signal Processing Pipeline

Before any volume conversion or ML inference, raw sensor readings pass through a three-stage signal processing pipeline implemented in Python. This pipeline mirrors the noise handling logic embedded in the ESP32-S3 firmware to ensure consistency between embedded and cloud-side processing.

5.1 Processing Stages

Stage	Name	Logic
1	Range Validation	Reject any reading outside [2.0 cm, 400.0 cm]. Readings below 2 cm indicate sensor obstruction; readings above 400 cm exceed tank height.

2	Spike Rejection	Reject any reading where the absolute change from the previous smoothed value exceeds 50 cm. This eliminates ultrasonic interference artefacts and sudden spurious echoes.
3	Weighted Smoothing	Apply a weighted moving average: smoothed = 0.3 × new + 0.7 × previous. This reduces measurement noise while preserving real level changes. First reading is accepted without smoothing.
4	Volume Conversion	Convert the smoothed distance to liquid volume in litres using the cylindrical geometry formula. Output is clamped to [0, TANK_CAPACITY_L].

5.2 Signal Processing Parameters

Parameter	Value	Purpose
SENSOR_MIN_CM	2.0 cm	Lower bound — rejects sensor obstruction
SENSOR_MAX_CM	400.0 cm	Upper bound — tank height ceiling
MAX_DELTA_CM	50.0 cm	Maximum allowed change per reading
SMOOTH_WEIGHT (new)	0.30	Weight for incoming reading in smoother
SMOOTH_WEIGHT (prev)	0.70	Weight for previous smoothed value

6. ML Feature Preparation

After volume conversion, the processed data is transformed into the same 17-feature input matrix used to train the Milestone 2 models. All label encoders and the StandardScaler are loaded directly from the .pkl files saved in Milestone 2, ensuring zero drift between training and inference.

6.1 Artifact Loading

- le_shift.pkl — LabelEncoder for shift_period (morning / afternoon / night)
- le_tank.pkl — LabelEncoder for tank_id (TANK_01 through TANK_05)
- le_dow.pkl — LabelEncoder for day_of_week (Monday through Sunday)
- le_risk.pkl — LabelEncoder for risk_level (CRITICAL / NORMAL / WARNING)
- scaler.pkl — StandardScaler fitted on Milestone 2 training data
- model_rf_regressor.pkl — Trained Random Forest Regressor
- model_rf_classifier.pkl — Trained Random Forest Classifier

6.2 Derived Feature Construction from Live Data

Several features in the 17-column input matrix cannot be directly read from ThingSpeak and are derived from the processed data:

Derived Feature	Derivation Method
fill_level_cm	volume_to_fill_cm(volume_L) using cylindrical ratio
fill_percentage	(volume_L / TANK_CAPACITY_L) × 100
headspace_cm	TANK_HEIGHT_CM − fill_level_cm
capacity_remaining_litres	TANK_CAPACITY_L − current_fill_litres
water_sensor_adc_raw	Derived as (fill_pct / 100) × 4095 where field6 is unavailable
consumption_per_interval_L	field7 (total_usage_raw) / NUM_ACTIVE_TANKS
time_to_empty_hours	volume_L / consumption_rate_Lps / 3600 (capped at 999)
day_index	Day-of-year from timestamp (strftime('%j'))

7. ML Inference Results

The trained Random Forest models from Milestone 2 are applied to the scaled live feature matrix to generate two outputs per sensor reading: a predicted fill level in litres (regression) and a risk classification label (classification).

7.1 Inference Pipeline

- Step 1: Load all .pkl artifacts (encoders, scaler, models).
- Step 2: Build processed DataFrame from raw ThingSpeak data (Step 5 pipeline).
- Step 3: Encode categorical columns using saved label encoders.
- Step 4: Scale feature matrix using saved StandardScaler.
- Step 5: Run Random Forest Regressor to predict current_fill_litres.
- Step 6: Run Random Forest Classifier to predict risk_encoded.
- Step 7: Decode risk label using le_risk.inverse_transform().
- Step 8: Compute derived output columns (predicted_fill_pct, prediction_error_L, alert_priority).

7.2 Output Columns Generated

Output Column	Description
predicted_fill_litres	RF Regressor output: predicted volume in litres
predicted_fill_pct	predicted_fill_litres as a percentage of TANK_CAPACITY_L
predicted_risk_level	RF Classifier output: CRITICAL / WARNING / NORMAL
prediction_error_L	predicted_fill_litres − current_fill_litres (residual)
alert_priority	Numeric alert rank: CRITICAL=3, WARNING=2, NORMAL=1
refill_needed_flag	Binary: 1 if fill_percentage ≤ 20%, else 0

7.3 Risk Classification Thresholds

Risk Level	Fill Threshold	Volume (approx.)	Recommended Action
CRITICAL	≤ 10%	≤ 5,027 L	Immediate refill required
WARNING	10.1% – 25%	5,027 – 12,566 L	Schedule refill soon
NORMAL	> 25%	> 12,566 L	No action required

8. Visualisation Dashboard

A six-panel Milestone 3 dashboard is generated and saved as milestone3_dashboard.png. Each panel presents a distinct aspect of the live prediction output to enable rapid operational assessment.

8.1 Dashboard Panels

Panel	Visualisation	Key Information Provided
1	Actual vs Predicted Fill (line plot)	Overlaid solid (actual) and dashed (predicted) fill level lines per tank over time. Confirms model tracking accuracy.
2	Predicted Fill % Over Time (line plot)	Fill percentage trends per tank with horizontal threshold lines at 10% (CRITICAL) and 25% (WARNING) for visual alert context.
3	Predicted Risk Distribution (bar chart)	Count of NORMAL, WARNING, and CRITICAL predictions across the entire live dataset. Bars are colour-coded green / amber / red.
4	Total Predicted Volume — All Tanks (area chart)	Summed predicted volume across all active tanks over time. Provides a plant-level inventory view.
5	Prediction Error Distribution (histogram)	Distribution of residuals (predicted − actual). A narrow, symmetric distribution centred at zero confirms model reliability.
6	Latest Fill % per Tank (bar chart)	Most recent predicted fill percentage per tank, colour-coded by risk level. Provides an at-a-glance tank status snapshot.

9. Live Status Console

A text-based live status console is printed at the end of the inference pipeline, showing the most recent reading and ML prediction for each active tank. The console is structured as follows:

```
============================================================
  LIVE TANK STATUS — LATEST READINGS + ML PREDICTIONS
  Timestamp: 2024-03-15 14:32:00
============================================================
 TANK_01
 Actual Fill      :   38,420.5 L  (76.4%)
 Predicted Fill   :   38,398.2 L  (76.3%)
 Prediction Error :      -22.3 L
 Time to Empty    :       18.42 hrs
 Risk Level       : 🟢 NORMAL
```

For CRITICAL tanks, the console appends: ⚠️ ACTION: Immediate refill required!
For WARNING tanks, the console appends: ⚠️ ACTION: Schedule refill soon.

10. Output CSV — milestone3_predictions.csv

All prediction outputs are consolidated into a structured CSV file for handoff to Milestone 4 Power BI dashboard development. The file contains one row per tank per timestamp, with 15 columns covering both sensor-derived and ML-predicted values.

10.1 CSV Schema

Column	Type	Description
timestamp	datetime	ISO 8601 timestamp from ThingSpeak entry
tank_id	string	TANK_01, TANK_02, or TANK_03
current_fill_litres	float	Sensor-derived liquid volume (post-processing)
fill_percentage	float	Fill level as % of tank capacity
predicted_fill_litres	float	RF Regressor prediction
predicted_fill_pct	float	Predicted fill as % of capacity
predicted_risk_level	string	CRITICAL / WARNING / NORMAL
alert_priority	integer	3 = CRITICAL, 2 = WARNING, 1 = NORMAL
time_to_empty_hours	float	Predicted hours until depletion at current rate
refill_needed_flag	integer	1 if fill ≤ 20%, else 0
consumption_per_interval_L	float	Litres consumed in current 15-second interval
consumption_rate_Lps	float	Instantaneous consumption rate in L/s
total_volume_all_tanks_L	float	Combined fill volume across all active tanks
sensor_status	string	OK / INVALID / SPIKE REJECTED / NO DATA
prediction_error_L	float	Residual: predicted − actual fill (litres)

11. Project Deliverables & Milestones

Milestones 1, 2, and 3 are now complete. The system has a fully operational IoT-to-ML pipeline producing live predictions.

M	Title	Key Deliverables	Status
1	System Architecture Design	Circuit design, pin mapping, ThingSpeak firmware	**Complete**
2	Dataset Generation & ML Model Development	Synthetic dataset, trained RF Regressor & Classifier	**Complete**

3	Live Data Simulation & Model Integration	ThingSpeak fetch, volume conversion, live ML predictions, prediction CSV	**Complete**
4	Analytics Output & Dashboard Visualization	Power BI dashboard with trends, risk alerts, and inventory KPIs	Upcoming
5	Result Analysis & Industry Interpretation	Final report, safety improvement demonstration, inventory insights	Upcoming

12. Conclusion

Milestone 3 successfully delivers a fully integrated live inference pipeline for the Smart Storage Tank Level and Usage Prediction System. Raw ultrasonic distance readings are fetched directly from the ThingSpeak cloud platform, passed through a physics-based volume conversion using cylindrical geometry (r = 2 m, h = 4 m), and cleaned using a three-stage signal processing pipeline before being fed into the trained Random Forest models.

The Random Forest Regressor accurately predicts the fill level in litres from live sensor features, while the Random Forest Classifier assigns a CRITICAL, WARNING, or NORMAL risk label to each reading in real time. All predictions are annotated with alert priority scores and exported to a 15-column structured CSV file, which is ready for ingestion into the Milestone 4 Power BI dashboard.

The end-to-end pipeline — from ESP32 sensor to ML prediction — now operates as a cohesive, production-grade system, validating the complete architecture design established at the outset of the project.